

Chiller Failure Incident Log & Root Cause Analysis

Jurong Hyperscale Campus · Data Centre · Singapore

Document ID	DC-02
Category	Incident & Failure Analysis
Classification	Maintenance
Date	2025-04
Prepared by	Critical Facilities Operations

1. Incident summary

At 02:40 on 14 March 2025, chiller CH-03 serving Data Hall 2 tripped on high motor winding temperature caused by a compressor bearing failure. The standby chiller CH-04 failed to start automatically because a BMS software update two weeks earlier had altered the sequencing logic. Rack inlet temperatures in Data Hall 2 rose from 24°C to 31.5°C within 14 minutes. Two customers' servers throttled performance, but no IT load was lost.

Time	Event
02:40	CH-03 trips; alarm raised in the NOC
02:44	CH-04 auto-start fails (sequencing fault)
02:49	On-call technician dispatched
02:54	Rack inlet temperature reaches 31.5°C
02:58	CH-04 started manually; temperatures begin to recover
03:35	Data Hall 2 back within SLA

Attendance: on-call technician Muhammad Hafiz (tel. 8765 4321); chiller OEM service engineer Ravi Kumar (NRIC S1234567D).

2. Root cause analysis (five whys)

- Why did CH-03 fail? The compressor bearing had worn progressively for about six weeks.
- Why was the wear not detected? Maintenance was time-based: quarterly inspection and oil analysis, the last of which took place ten weeks before the failure. No continuous vibration monitoring was installed.
- Why did the standby chiller not start? A BMS software update changed the sequencing logic, and failover was not re-tested after the update.
- Why was it not re-tested? The change process did not require failover testing after control-software changes.

3. Contributing factors

- There was no alarm for 'standby chiller not ready'.
- Failover had last been tested four months earlier.

4. Corrective and preventive actions

- Tri-axial vibration sensors installed on all chiller compressors and condenser-water pumps, with machine-learning anomaly detection on bearing defect frequencies.
- Automated monthly failover test of every chiller, with results logged.
- Change control: any BMS or controls software change now requires a documented failover retest before sign-off.
- Critical spare bearing kits held on site.

5. Results after 12 months

There were no unplanned chiller trips in the 12 months after these actions. Vibration monitoring raised three early warnings - a bearing defect, a shaft misalignment and pump cavitation - each detected two to five weeks before it would have caused a failure and repaired during a planned maintenance window.